Irabiel Romero

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Research Interests

General:	Mathematical modeling, numerical analysis, large-scale inverse problems, PDE-constrained optimization, Bayesian inference, uncertainty quantification, optimal experimental design, optimization under uncertainty, numerical optimization, numerical linear algebra, multigrid solvers, scientific machine learning
Applications:	Power-grid modeling

Education

Dec. 2026	**Ph.D. Applied Mathematics**, University of California, Merced, CA, USA (expected). Thesis: *Scalable Methods for Estimation of Power Systems under Uncertainty* (Advisors: Prof. Noemi Petra and Prof. Roummel Marcia)
May 2021	**M.S. Applied Mathematics**, California State University, Long Beach, CA, USA. Thesis: *Numerical Methods for Multiscale Modeling in Bacterial Colony Growth* (Advisor: Prof. Hui Sun)
May 2018	**B.S. Applied Mathematics** and **B.A. Physics**, California State University, Long Beach, CA, USA
May 2016	**A.S. Mathematics**, Bakersfield College, Bakersfield, CA, USA

Ph.D.'s Research Experience

- **Graduate Research Intern**, Center for Applied Scientific Computing (CASC), Lawrence Livermore National Laboratory, Livermore, CA, June 2026–Aug. 2026
  - Integrated a machine learning surrogate into the transient stability-constrained AC optimal power flow (TS-ACOPF) solver, replacing stability constraints that are computationally intractable to enforce directly.
  - Designed and ran validation tests comparing surrogate-constrained solutions against full simulation.
- **Graduate Student Researcher**, Lawrence Livermore National Laboratory, Livermore, CA, Spring 2025–Spring 2026 (Project manager: Dr. Cosmin Petra)
  - Extended LLNL's security-constrained AC optimal power flow (ACOPF) solver, ExaJuGO, to analyze wildfire-driven N-k contingencies on the California Test System, and developed interactive visualization tools for regional and system-wide analysis.
  - Developed a convolutional neural network (CNN) surrogate for generator stability and integrated it into ACOPF to enforce transient stability constraints that would otherwise be computationally intractable.
- **Data Science Summer Institute**, Lawrence Livermore National Laboratory, Livermore, CA, May 2024–Aug. 2024 (Project manager: Dr. Ignacio Aravena)

  - Developed a general parallel framework (C++, MPI, HDF5) using Particle Swarm Optimization to generate DC power flow databases for studying the effects of line susceptance on power flows and voltage angles.
  - Developed CNN models to classify heartbeats from electrocardiography (ECG) signals, identify the type of irregularity, and estimate cardiac activation times.

- **Data Science Challenge Internship**, Lawrence Livermore National Laboratory, Livermore, CA, July 2023–Aug. 2023 (Project manager: Dr. Mikel Landajuela; project support: Brian Gallagher and Cindy Gonzales)
  - Served as a team lead in developing a 1D CNN that reconstructs the heart's transmembrane voltages and activation times from ECG signals.

- **Graduate Student Researcher**, University of California, Merced, CA, Jan. 2023–current (Advisors: Prof. Noemi Petra and Prof. Roummel Marcia)
  - Developed a Bayesian inverse problem governed by the DC power flow equations to estimate voltage angles and uncertain line parameters, and an optimal experimental design method for PMU placement.
  - Led a project with Dr. Cosmin G. Petra (CASC) and Dr. Kathleen Schmidt (Uncertainty Quantification group lead), LLNL, Fall 2026: developed an ML surrogate for TS-ACOPF that learns the distribution of transient stability index (TSI) values over all possible bus fault locations and uses it to estimate the probability that the system becomes unstable; added conformal prediction to give calibrated, distribution-free bounds on that probability.

- **Graduate Student Researcher**, University of California, Merced, CA, May 2022–Aug. 2022 (Advisors: Prof. Noemi Petra and Prof. Maxime Theillard)
  - Developed a generic algebraic multigrid solver for M-matrices, and an inverse problem governed by an advection–diffusion equation to recover the diffusion parameter from sparse data.

Master's Research Experience

- **Graduate Researcher**, California State University, Long Beach, CA, May 2020–May 2021 (Advisor: Prof. Hui Sun)
  - Extended the modified Newton–multigrid solver to a 2D reaction–diffusion model of bacterial crossfeeding, and coupled it with a kinetic Monte Carlo model for agent-based cell simulation.

- **Graduate Researcher**, California State University, Long Beach, CA, June 2019–Aug. 2019 (Advisor: Prof. Hui Sun)
  - Developed a modified Newton–multigrid solver for a stiff system of nonlinear reaction–diffusion partial differential equations (PDEs) describing chemical concentrations in a bacterial colony.

- **Graduate Researcher**, University of California, San Diego, CA, June 2018–Aug. 2018 (Advisor: Prof. Hui Sun)
  - Extended a 3D computational bacterial colony model with PDEs for additional chemical species that influence cell metabolism, movement, and colony morphology.

Publications

- **I. Romero**, N.-Y. Chiang, I. Aravena, R. Marcia, N. Petra, and C. G. Petra. "Real-time power system machine learning-assisted transient stability preventive control". In preparation.
- **I. Romero**, H. Kannan, A. Valdez, M. Winger, B. Li, and H. Sun. "Advanced numerical methods for reaction-diffusion equations modeling the growth of bacterial colonies". SSRN preprint (2026)
- **I. Romero**, N.-Y. Chiang, I. Aravena, R. Marcia, N. Petra, J.-P. Watson, and C. G. Petra. "Wildfire-driven N-k contingency analysis for security-constrained AC optimal power flow in the California power grid". IEEE Electrical Power and Energy Conference (EPEC) (2026)
- **I. Romero**, R. Marcia, I. Aravena, and N. Petra. "Optimal PMU placement for state estimation with grid parameter uncertainty". 2025 IEEE PES Grid Edge Technologies Conference and Exposition (Grid Edge), IEEE (2025)
- A. Ranganath, **I. Romero**, M. Singhal, and R. Marcia. "Quasi-Adam: Accelerating Adam using quasi-Newton approximations". 2024 International Conference on Machine Learning and Applications (ICMLA), IEEE (2024)
- **I. Romero**. "Numerical methods for multiscale modeling in bacterial colony growth". Master's Thesis, California State University, Long Beach (2021)

Research Support

2025	LLNL subcontract, Lawrence Livermore National Laboratory (PI: Dr. Cosmin G. Petra; with Dr. Nai-Yuan Chiang), Spring–Summer 2025
2023–2026	Graduate Student Researcher, NSF CAREER #1654311 and NSF AMPS #229495
2022	Summer Research, Fields Family Fellowship and Applied Math Graduate Program Summer Research Fellowship, Summer 2022

Teaching Experience

Fall 2022	**Teaching Assistant**	**Math 131** - *Numerical Methods for Scientists and Engineers*, University of California, Merced
Spring 2022	**Teaching Assistant**	**Math 12** - *Calculus II*, University of California, Merced
Fall 2021	**Teaching Assistant**	**Math 24** - *Linear Algebra and Differential Equations*, University of California, Merced
Fall 2020	**Teaching Assistant**	**Math 112** - *Essential Algebra A*, California State University, Long Beach

Oral Presentations

- *Scalable Methods for Estimation of Power Systems under Uncertainty*, Optimization Seminar, University of California, Merced, CA, Fall 2026

- *Real-Time Power System Machine Learning-Assisted Transient Stability Preventive Control*, Optimization Seminar, University of California, Merced, CA, Spring 2026
- *Wildfire-driven N-k contingency analysis for security-constrained AC optimal power flow in the California power grid*, Optimization Seminar, University of California, Merced, CA, Fall 2025
- *Optimal PMU Placement for State Estimation with Grid Parameter Uncertainty*, Optimization Seminar, University of California, Merced, CA, Spring 2025
- *Optimal PMU Placement for State Estimation with Grid Parameter Uncertainty*, 28th Annual CASIS Workshop, Livermore, CA, Summer 2024
- *Embedding Power Flow into Machine Learning for Parameter and State Estimation*, Optimization Seminar, University of California, Merced, CA, Fall 2022
- *Numerical Methods for Multiscale Modeling in Bacterial Colony Growth*, 32nd Annual CSULB Student Research Competition, Long Beach, CA, Spring 2020
- *Spatiotemporal Establishment of Growing Bacterial Colonies*, 31st Annual CSULB Student Research Competition, Long Beach, CA, Spring 2019
- *Spatiotemporal Establishment of Growing Bacterial Colonies*, UCSD Q-Bio Seminar, San Diego, CA, Summer 2018

Poster Presentations

- *Real-Time Power System Machine Learning-Assisted Transient Stability Preventive Control*, 3rd SIAM Northern and Central California Sectional Conference (**NCC26**), Davis, CA, Fall 2026 (Poster Award)
- *Wildfire-driven N-k contingency analysis for security-constrained AC optimal power flow in the California power grid*, 2nd SIAM Northern and Central California Sectional Conference (**NCC25**), Berkeley, CA, Fall 2025 (Poster Award)
- *Optimal PMU Placement for State Estimation with Grid Parameter Uncertainty*, IEEE PES Grid Edge Technologies Conference and Exposition, San Diego, CA, Spring 2025
- *Optimal PMU Placement for State Estimation with Grid Parameter Uncertainty*, 28th Annual CASIS Workshop, Livermore, CA, Summer 2024
- *Hearts with Machine Learning*, LLNL Data Science Challenge, Livermore, CA, Summer 2023
- *Numerical Methods for Multiscale Modeling in Bacterial Colony Growth*, Graduate Research Conference, Long Beach, CA, Fall 2019
- *Spatiotemporal Establishment of Growing Bacterial Colonies*, Graduate Research Conference, Long Beach, CA, Fall 2018

Selected Honors and Awards

2026 Outstanding Poster Award, SIAM NCC26, Davis, CA
2025 Best Poster Award, SIAM NCC25, Berkeley, CA
2021 Fields Family Fellowship, UC Merced
2021 Outstanding Thesis Award, CSU Long Beach
2020 Richard Green Graduate Fellowship, CSU Long Beach
2019 Summer Student Research Assistantship, CSU Long Beach
2019 Samuel Councilman Scholarship, CSU Long Beach
2018 President's List, CSU Long Beach
2017 Dean's List, CSU Long Beach

Professional Development

- *Uncertainty Quantification Summer School*, University of Southern California, Los Angeles, CA, August 2026
- *Structure-Preserving Scientific Computing and Machine Learning Summer School and Hackathon* (PIMS CRG), University of Washington, Seattle, WA, June 2025
- *FEniCS 2022 Conference*, University of California, San Diego, La Jolla, CA, August 2022

Graduate Coursework

UC Merced:	Numerical Optimization (Fall 2023), Scientific Computing (Fall 2022), Advanced Methods of Applied Mathematics (Spring 2022), Numerical Solution of Differential Equations I and II (Fall 2021, Spring 2022), Partial Differential Equations (Spring 2022), Computational Inverse Problems (Fall 2021), Advanced Ordinary Differential Equations (Fall 2021)
CSU Long Beach:	Finite Element Method (Spring 2021), Mathematical Modeling with Graph Theory (Spring 2020), Sobolev Spaces (Fall 2019), Partial Differential Equations (Spring 2019), Numerical Analysis (Spring 2019), Nonlinear Differential Equations (Fall 2018), Numerical Linear Algebra (Fall 2018), Mathematical Modeling with Stochastic Differential Equations (Spring 2018)

Technical and Computational Skills

Languages:	Python, C++, MATLAB, Julia
Numerical tools:	FEniCS, hIPPYlib, IPOPT, PyTorch, TensorFlow, JAX, SciPy
Parallel programming:	OpenMP, MPI
Other:	Git, LaTeX, Vim